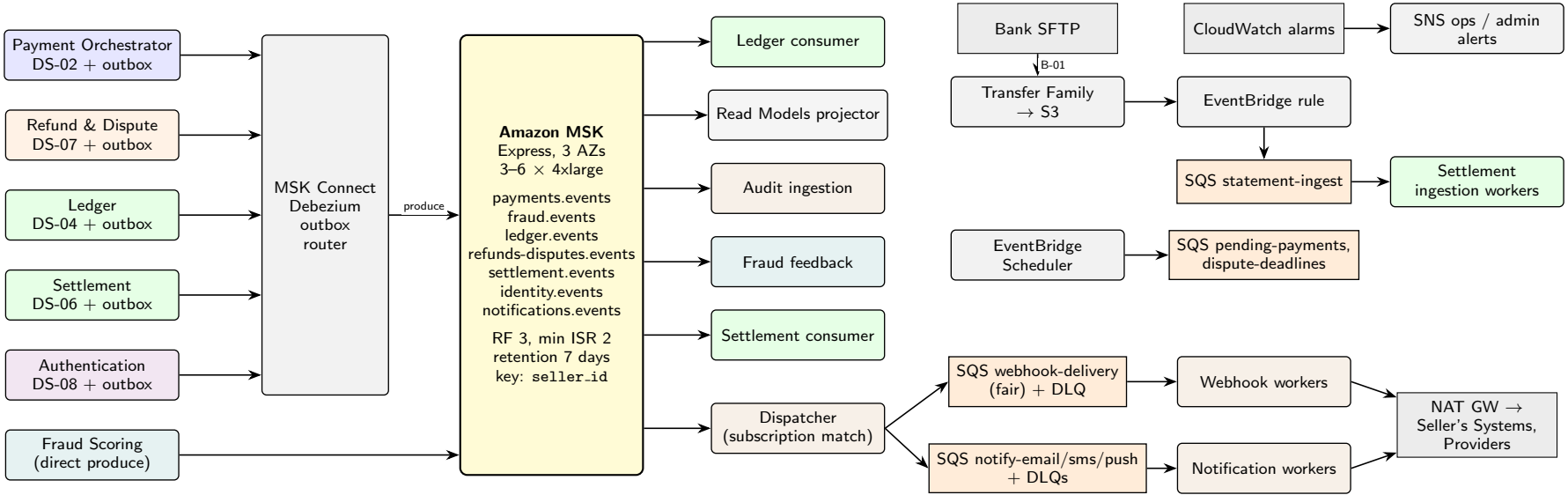
Payment Orchestrator
DS-02 + outbox
Refund & Dispute
DS-07 + outbox
Ledger
DS-04 + outbox
Settlement
DS-06 + outbox
Authentication
DS-08 + outbox
Fraud Scoring
(direct produce)
MSK Connect
Debezium
outbox
router
produce
Amazon MSK
Express, 3 AZs
3–6 × 4xlarge
payments.events
fraud.events
ledger.events
refunds-disputes.events
settlement.events
identity.events
notifications.events
RF 3, min ISR 2
retention 7 days
key: seller_id
Ledger consumer
Read Models projector
Audit ingestion
Fraud feedback
Settlement consumer
Dispatcher
(subscription match)
Bank SFTP
B-01
Transfer Family
→ S3
CloudWatch alarms
SNS ops / admin
alerts
EventBridge rule
SQS statement-ingest
Settlement
ingestion workers
EventBridge
Scheduler
SQS pending-payments,
dispute-deadlines
SQS webhook-delivery
(fair) + DLQ
Webhook workers
SQS notify-email/sms/push
+ DLQs
Notification workers
NAT GW →
Seller's Systems,
Providers